ALGEBRA 1 - UNIT 1

Section 1.5
Equations as Models

Using equations to represent a situation, not just to get an answer.

> **Mathematical idea:** A model is useful when the equation, the solution, and the situation all make sense together.

Today we will write equations from situations, solve them, and explain what the solution means in context.

Learning Target

Students will write, solve, and interpret equations that model situations.

I can statements

- I can write equations to represent situations.
- I can solve equations and interpret solutions.
- I can determine whether a solution is reasonable.
- I can explain what a solution means in context.
- I can compare equation solving with other solution strategies.

What's To Come

This is a long-range preview for future lessons. Do not solve it today. Instead, annotate it individually. Circle anything familiar, underline symbols or directions that seem important, and write notes about what information you think will be needed later.

As you annotate, ask yourself: What decision is the equation helping someone make? What does the crossing point on the graph seem to represent? What information would you need before choosing a plan? How could a mathematically correct answer still need to be interpreted carefully?

Future Task

A community center is comparing two membership plans. Both plans charge a starting fee and a monthly cost, but the plans are built differently.

Plan A

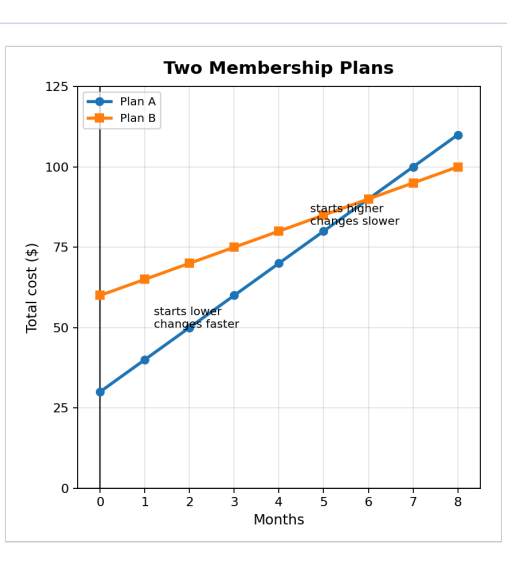
Starts with a lower fee but costs more each month.

$$C = 10m + 30$$

Plan B

Starts with a higher fee but costs less each month.

$$C = 5m + 60$$



1. Use evidence to decide when the plans have the same cost.
2. Explain what that solution means in the situation.
3. Decide which plan is better for a short membership and which is better for a long membership.
4. Explain why solving an equation is only part of making a real-world decision.

Intro

Directions

Read the introduction aloud with a partner or in a small group. As you read, annotate the text by highlighting important ideas, circling key vocabulary, and writing short notes about connections, questions, or predictions in the margins.

An equation can be more than a math sentence. It can be a **model**. A model uses math to represent something happening in a situation.

For example, if a ticket booth charges a starting fee and then adds the same amount for each ticket, an equation can show the total cost. The numbers in the equation have meaning. One number might show the starting cost. Another number might show how much the cost changes each time.

Solving an equation gives a value that makes the equation true. In a situation, that value should also answer a question. If the answer is 6, we should know whether that means 6 tickets, 6 months, 6 dollars, or something else.

A good model must be checked. The equation should match the story. The solution should be reasonable. The explanation should connect back to the original situation.

Stop and Discuss

1. How is an equation model different from an equation with no context?
2. Why is it not enough to only say the numerical answer?
3. What should you check before deciding a solution is reasonable?

Equations Match the Story

When we write an equation from a situation, we first decide what the variable represents. Then we connect the numbers in the story to the parts of the equation.

> **Equation model:** An equation that represents a situation and helps answer a question about that situation.

Example 1

A taxi charges a starting fee of $4 plus $2 for each mile. Write an equation for the total cost after m miles. Then explain what each part of the equation means.

You Try It 1

A school dance charges $6 to enter plus $3 for each snack ticket. Write an equation for the total cost after t snack tickets. Explain what the variable and each number mean.

Stop and Discuss

1. Which number in Example 1 is the starting amount?
2. Which number tells the change for each mile?

Solving Should Answer the Question

Solving gives the value that makes the equation true. Interpreting explains what that value means in the situation.

Modeling check: What does the variable represent? What operation is happening? What question does the solution answer?

Example 2

A gym charges $20 to join and $15 each month. The total cost is $80.

1. Write an equation to represent the situation.
2. Solve the equation.
3. Explain what the solution means.

You Try It 2

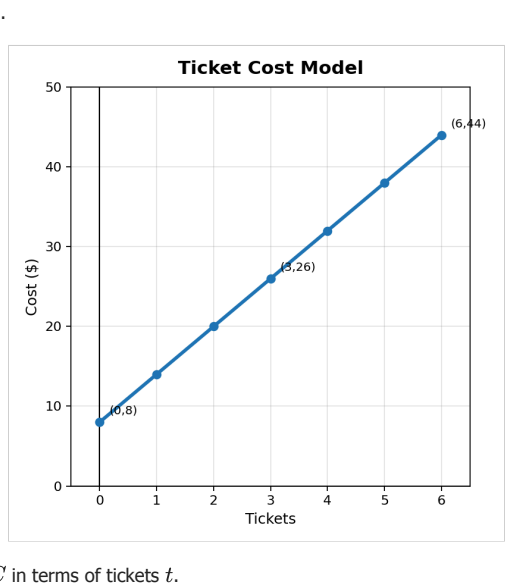
A club has $18 already saved and earns $6 for each car washed. The club wants $60 total. Write and solve an equation, then explain the solution in context.

Graphs Can Help Us Read an Equation Model

A graph shows the same model visually. The starting value appears where the graph begins at input 0. The change shows how the graph rises or falls as the input increases.

Example 3

The graph shows a ticket cost model.



1. Write an equation for the cost C in terms of tickets t.
2. Use the graph or equation to find the cost of 5 tickets.
3. Explain what the starting value means.

You Try It 3

A table shows a total cost of $7 at 0 items, $12 at 1 item, $17 at 2 items, and $22 at 3 items. Write an equation model and explain what the 7 and the 5 mean.

Stop and Discuss

1. How does the graph show the same starting value as the equation?
2. Why can a graph help check whether an equation model is reasonable?

A Correct Equation Can Still Need a Reality Check

After solving, we should ask whether the answer makes sense. Some contexts need whole numbers. Some answers may be too large, too small, or have the wrong unit for the situation.

Example 4

A class needs $50 for supplies. Each student contributes $8. A student writes and solves $8s = 50$ and gets $s = 6.25$.

1. Explain what the solution 6.25 means mathematically.
2. Explain why the solution may not make sense as a final answer in this context.
3. Decide what the class should do and justify your answer.

You Try It 4

A group needs $70 for a trip. Each fundraiser ticket earns $12. Solve $12t = 70$. Then explain why the exact solution may need to be adjusted in the real situation.

Stop and Discuss

1. When might a decimal answer be reasonable?
2. When might a decimal answer need to be rounded or rejected?

Summary

- An equation model represents a situation with symbols and numbers.
- The variable should have a clear meaning.
- Numbers in the equation should connect to parts of the story.
- Solving an equation gives a value that makes the equation true.
- Interpreting the solution explains what the value means in context.
- A reasonable answer should fit the situation and use the correct unit.
- Tables and graphs can help check whether an equation model makes sense.

Common Mistakes

- **Writing an expression instead of an equation:** An equation has an equal sign and often answers a specific question.
- **Not defining the variable:** A solution is hard to interpret if the variable was never explained.
- **Swapping the starting value and rate:** The starting amount and repeated change have different meanings.
- **Stopping after solving:** A model needs an interpretation, not just a number.
- **Ignoring units:** The answer might represent miles, tickets, months, people, dollars, or another quantity.
- **Forgetting the reality check:** Some mathematical answers need to be adjusted because of the context.

Optional Future Task Reflection

Optional Reflection: Look back at the future task. Do not solve it yet.

1. Highlight one part that makes more sense now than it did at the beginning.
2. Circle one part that still feels unfamiliar.
3. Write one sentence about what an equation solution would mean in that situation.